

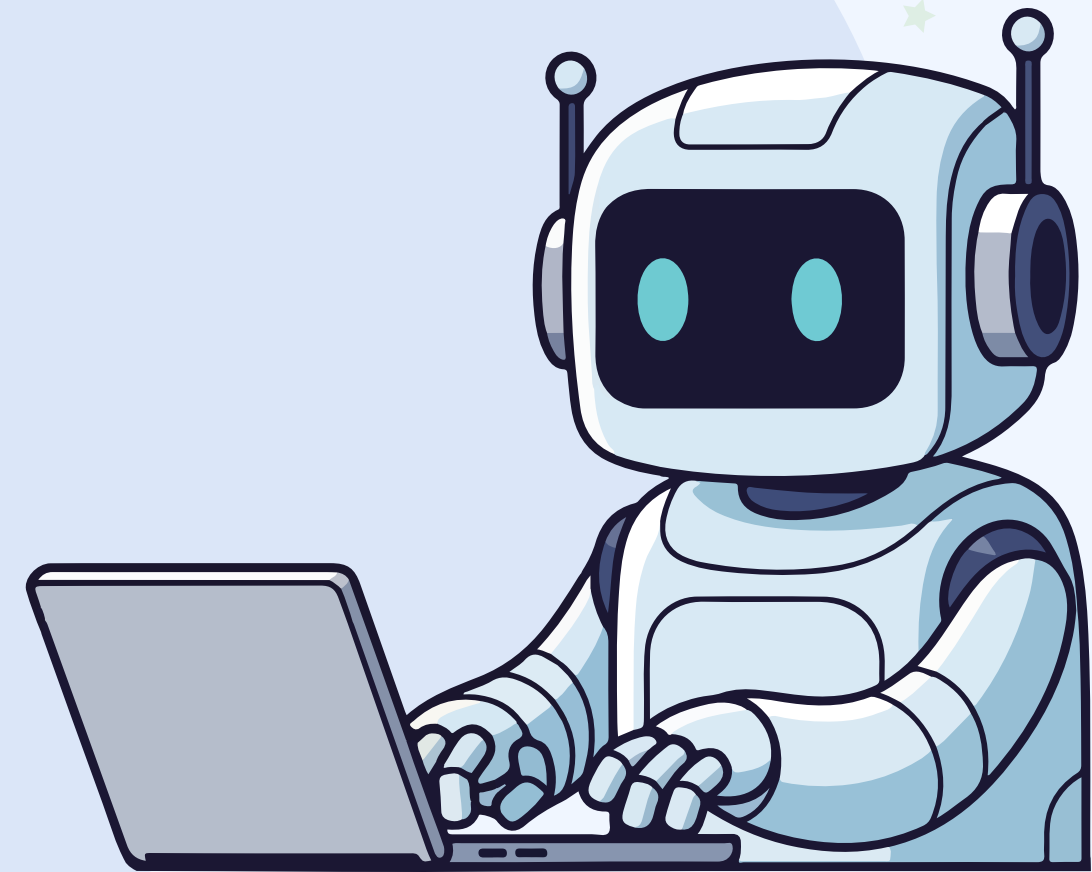
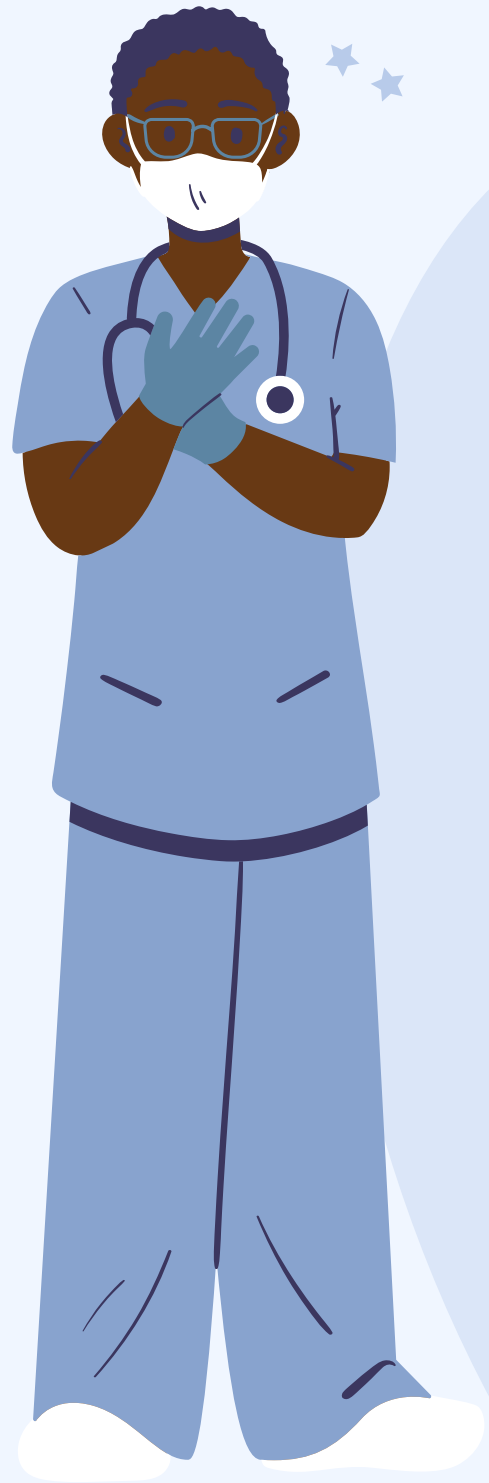
group 3

diabetes risk prededction



Problem

- Diabetes is a serious chronic disease
- Often detected late
- Late detection leads to complications
- Need for early risk prediction

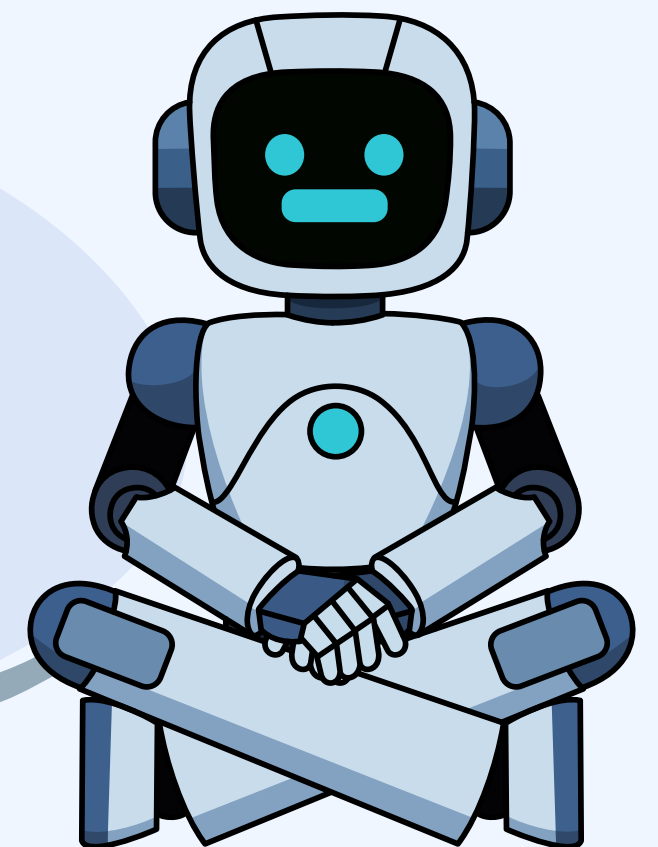


Objectives

**Build a machine
learning model**

**Predict diabetes
risk levels**

**Use health and
lifestyle features**



Project overview

- Dataset: 6000 records
- Features: medical & lifestyle data
- Models: Logistic Regression, SVM, Random Forest
- Goal: classify individuals into risk categories



Exploratory Data Analysis (EDA)

Key insights From EDA

Main Findings

Class Distribution: Most cases fall within the Low Risk category.

Main Findings

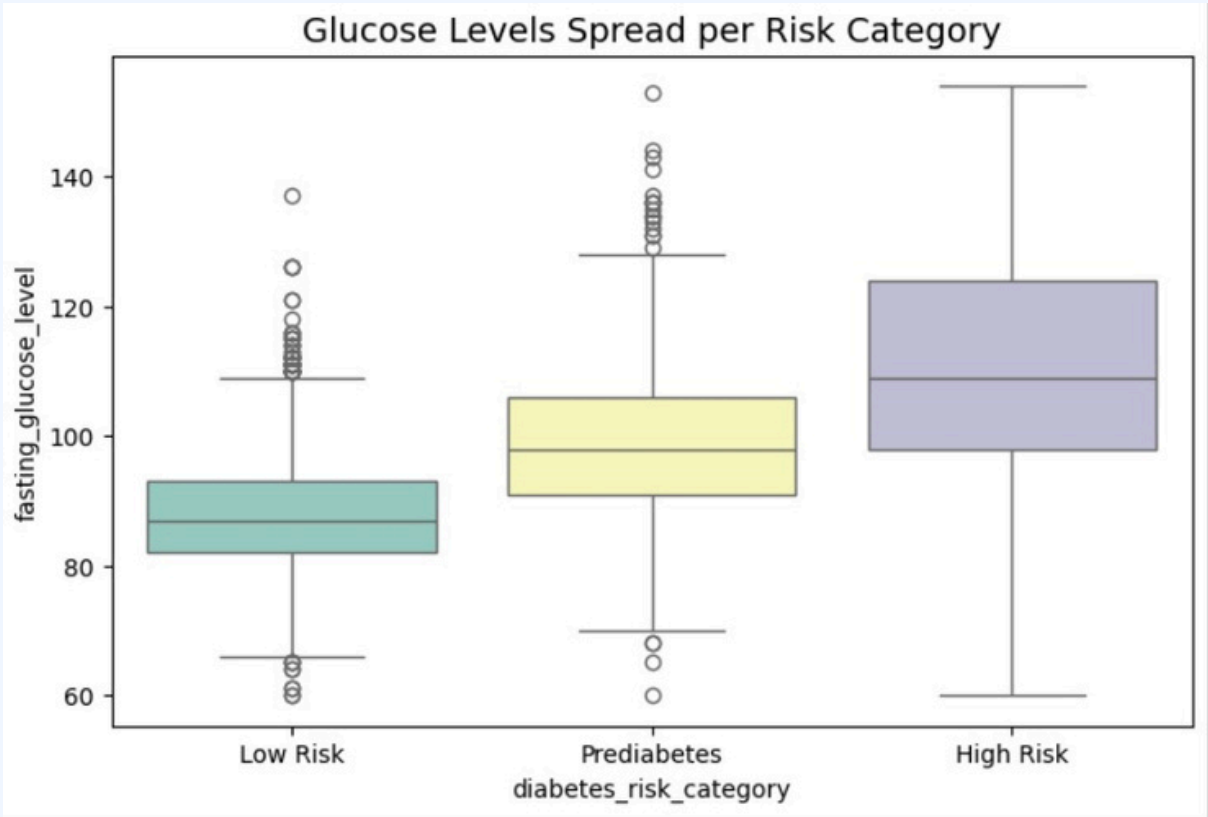
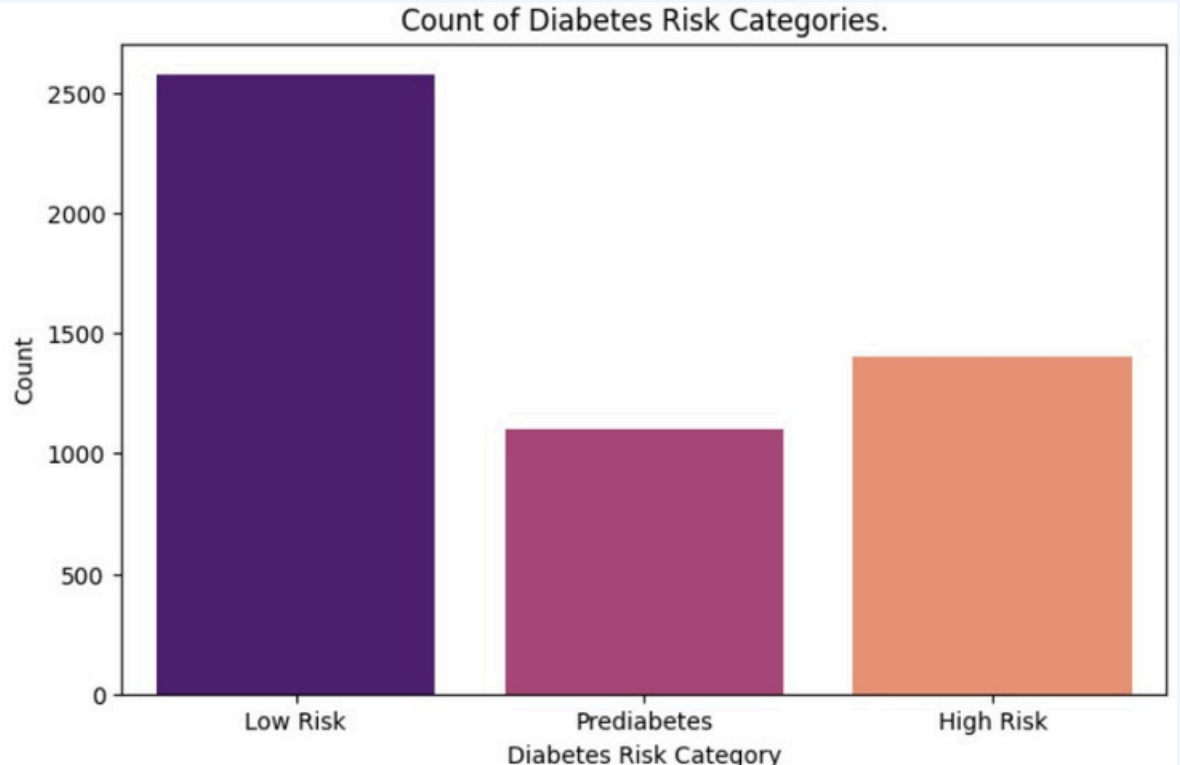
Data Balance: A slight class imbalance was observed within the dataset.

Key Observations

Outliers: The Box plot analysis revealed the presence of outliers in the glucose data that require handling

Key Observations

Data Preparation: Based on the EDA, it is essential to perform Data Preprocessing steps before building the prediction model



Key insights From EDA

Overview 01

Diverse age range participants are aged between 20 and 80

Overview 02

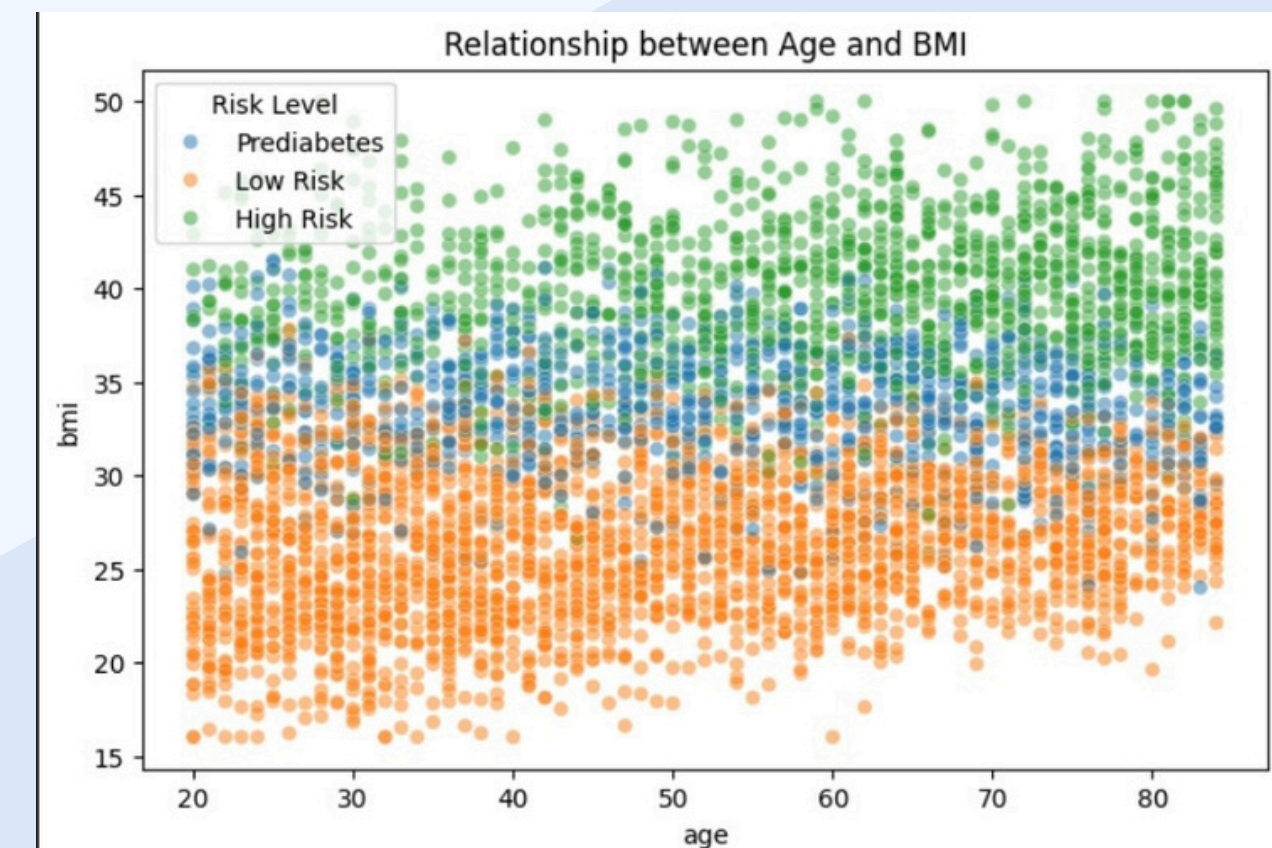
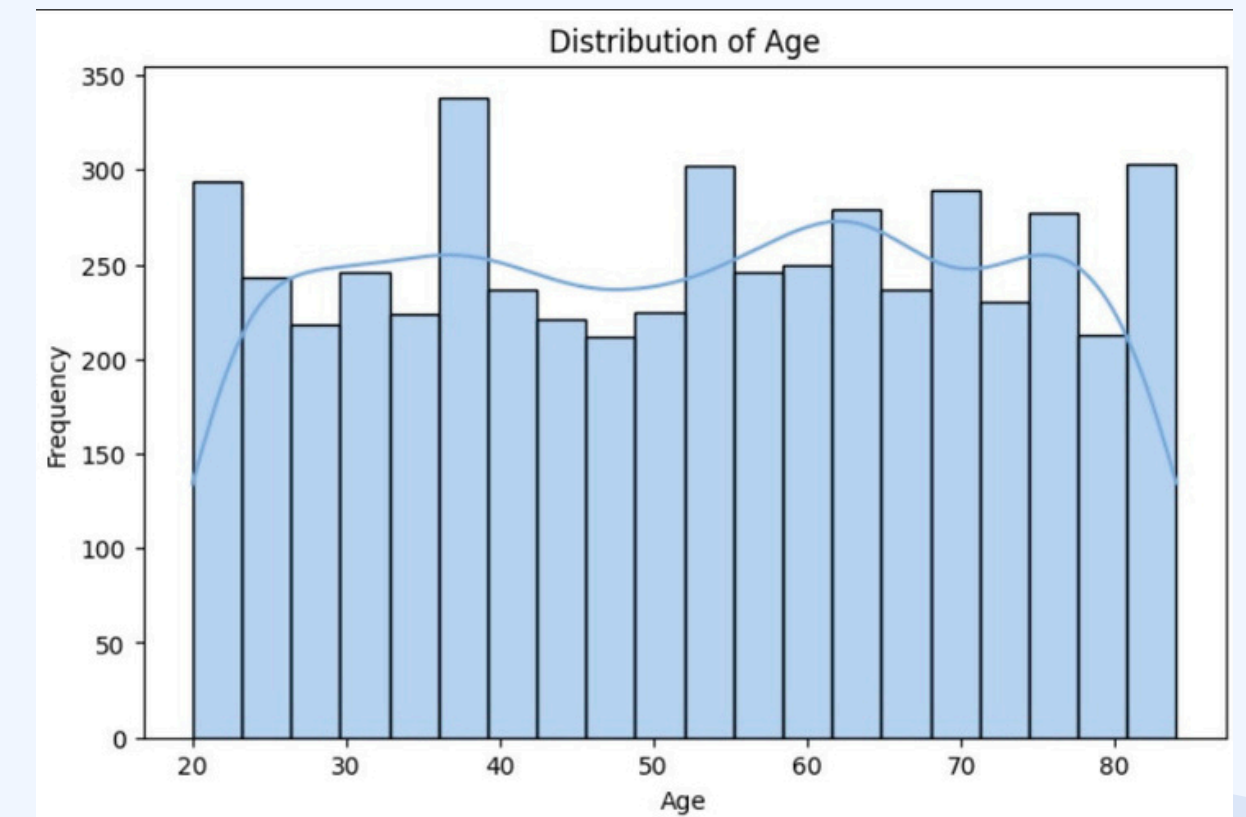
ensureins the model is not biased toward a specific age range

Overview 03

Higher BMI values are mor closely associated with High Risk
Risk levels across all ages

Overview 04

BMI is a critical factor for our prediction model



Model Performance

Model Performance

Models used:

- Logistic Regression
- SVM
- Random Forest

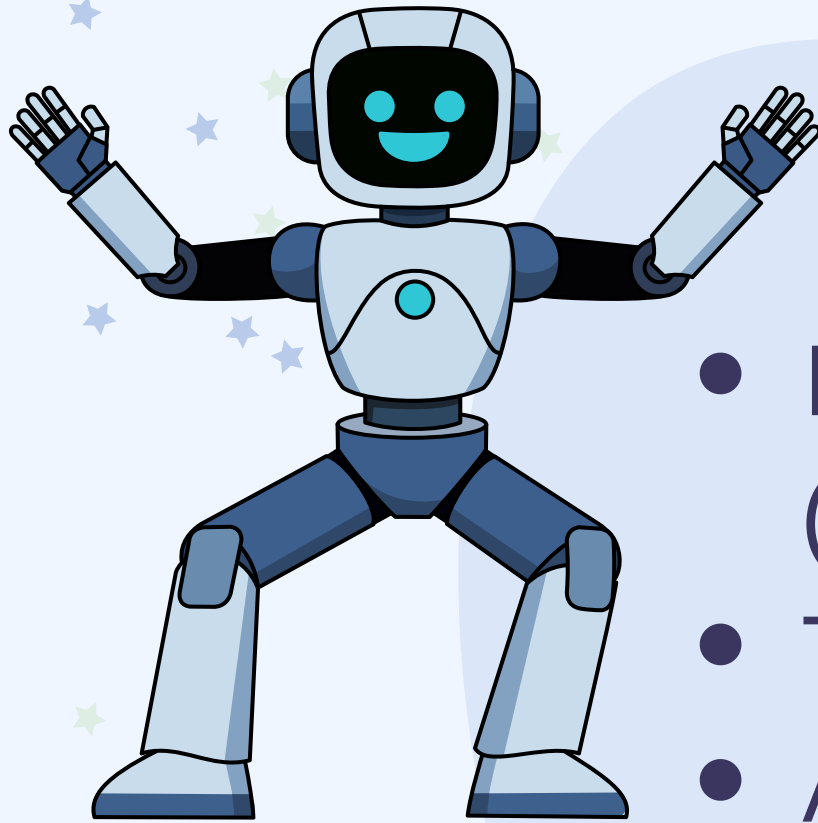
Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score

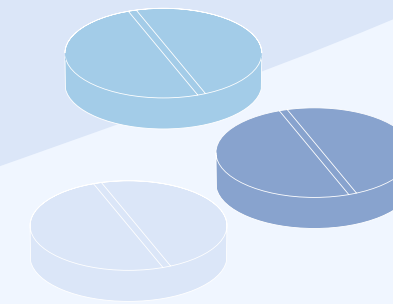
model	accuracy	precision	recall	F1-score
Logistic regression	0.81	0.80	0.81	0.72
SVM	0.91	0.92	0.91	0.91
Random forest	0.91	0.91	0.91	0.91

- SVM achieved the highest accuracy (~0.915)
- Random Forest showed similar performance (~0.913)
- Logistic Regression had lower performance
- Fixing data leakage by removing diabetes_risk_score improved model reliability
- Random Forest was selected as the final model

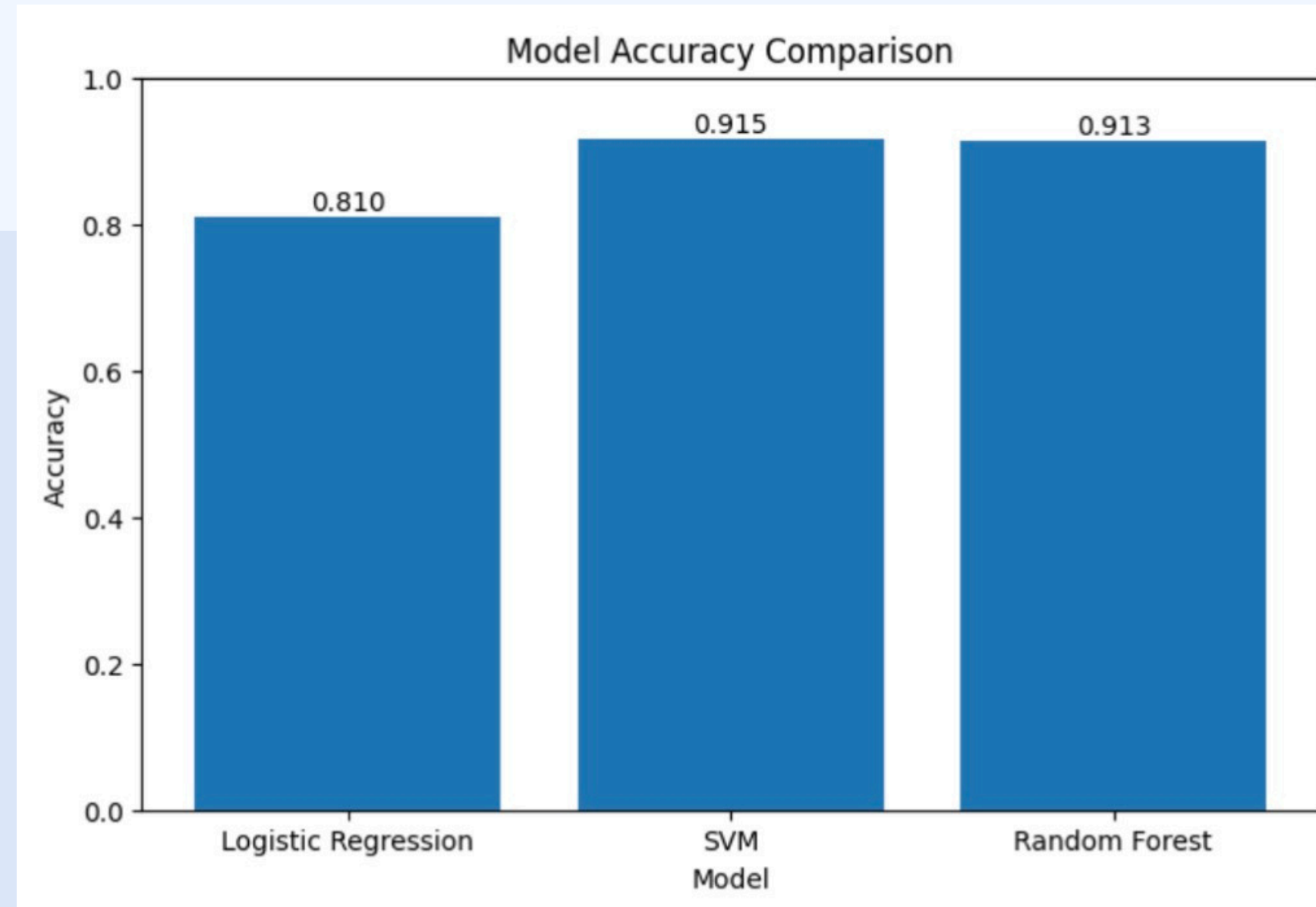
Findings



- Random Forest achieved **100% accuracy** (unrealistic result)
- This indicates **high overfitting**
- As mentioned earlier, this was due to **data leakage**
- After removing the feature, results became more realistic



Findings



- Models achieved strong performance (~91% accuracy)
- Performance was consistent across models
- Models captured real patterns (Glucose & BMI)
- Useful for early diabetes risk prediction



Limitations & Future Work

Limitations

- Class imbalance in the dataset
- Limited number of features
- May not generalize well to new data

Future Work

- Hyperparameter tuning
- Handle class imbalance (e.g., SMOTE)
- Use more comprehensive datasets





Thank you for your
attention

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